

Secure Link

Privacy-Preserving Record Linkage with XGBoost

Project Highlight

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Project Overview

SecureLink is a privacy-preserving record linkage (PPRL) system that matches records belonging to the same person across two separate datasets without exposing the underlying sensitive data. It was built for a healthcare context, where two parties may hold records about the same patient but cannot legally share raw clinical text with one another.

The system lets the data be matched in an encoded, noise-protected form rather than in the clear, using Bloom filter encoding and Local Differential Privacy (LDP). Five linkage models were explored across the team; this document covers the XGBoost component, which I designed, built, integrated, and evaluated.

My Contribution

I owned the XGBoost prototype end to end and trained it as a binary classifier (match vs non-match) under three configurations, to measure the real cost of adding privacy:

- **Raw baseline.** Text converted to features with TF-IDF, reduced to 200 dimensions with Truncated SVD. No privacy applied, used as the performance benchmark.
- **Local Differential Privacy (LDP).** Calibrated random noise added to the feature vectors, with noise strength controlled by a privacy budget (ϵ).
- **Bloom filter encoding.** Records hashed into bit vectors and compared with Dice similarity, using that similarity score as the model input.

I also integrated the model into the team's React dashboard so results could be run and viewed interactively.

Key Results

Measured on the real project dataset ($\epsilon = 7$). Adding LDP privacy cost only about 1.7% in F1 versus the raw baseline, showing strong privacy protection at minimal accuracy loss. The Bloom filter approach traded more recall for its privacy guarantees.

Configuration	F1	Accuracy	Precision	Recall
Raw XGBoost (baseline)	0.9118	0.9131	0.9237	0.9003
LDP XGBoost ($\epsilon = 7$)	0.8960	0.8980	0.9118	0.8808
Bloom Filter XGBoost ($\epsilon = 7$)	0.7252	0.8837	0.8754	0.6189

Real project results. Source: project documentation (non-confidential).

Demonstration (Synthetic Data)

To illustrate the approach safely, the pipeline was re-run on randomly generated dummy data that contains no real records and no personal information. The pattern matches the real project: each privacy step costs some performance, and the Bloom filter loses the most on recall.

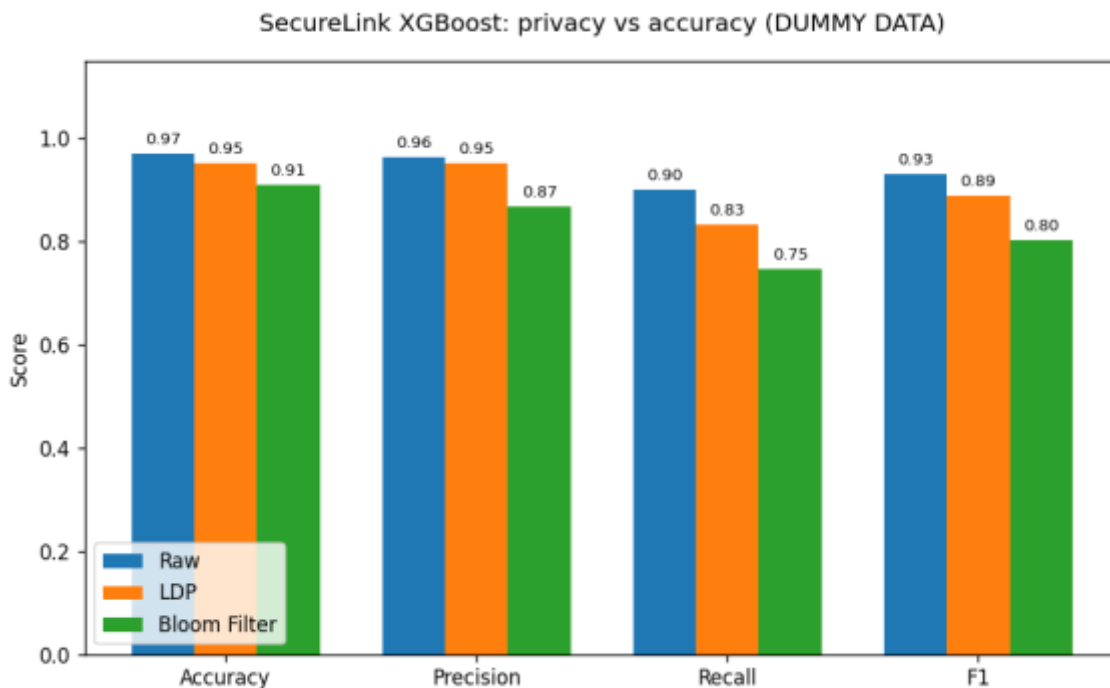


Figure 1. Model comparison across the three privacy configurations (synthetic data).

SecureLink: sample matches and non-matches (DUMMY DATA)

Type	Note A (text1)	Note B (text2)	Dice sim
Match	fracture chronic with stable review p...	presents prescribed review with patie...	0.894
Match	chronic acute reflux chronic follow i...	acute ibuprofen chronic follow reflux...	0.878
Match	prescribed of chronic with eczema of ...	salbutamol eczema prescribed mild his...	0.897
Match	ventolin patient history presents mig...	patient with presents history ventoli...	0.864
Match	with mild patient mild acute patient ...	asthma patient prescribed patient acu...	0.872
Non-match	aspirin presents patient up prescribe...	acute salbutamol chronic prescribed r...	0.763
Non-match	chronic with with patient history pre...	history presents prescribed with pres...	0.804
Non-match	chronic atorvastatin with of of prese...	up review patient paracetamol mild of...	0.786
Non-match	patient history prescribed of paracet...	prescribed patient with chronic prese...	0.781
Non-match	prescribed eczema with prescribed sal...	history mild patient prescribed stabl...	0.743

Figure 2. Sample matched and non-matched record pairs with Dice similarity scores (synthetic data).

Technical Notes

Details an interviewer typically asks about an ML classifier:

- **Task:** binary classification, 2 classes (match / non-match).
- **Class imbalance:** record-pair data is heavily skewed toward non-matches. For the Bloom filter model, non-matches were downsampled to a 3:1 ratio against matches before training, and the class distribution was logged before and after.
- **Features:** 200 SVD components for the raw and LDP models; a single Dice-similarity feature for the Bloom filter model.
- **Validation:** 70/30 train/test split with a fixed random seed; evaluated on Accuracy, Precision, Recall, and F1.
- **Hyperparameters:** XGBoost defaults were used as a deliberate baseline. A next iteration would tune `scale_pos_weight` (for the imbalance), `max_depth`, `learning_rate`, and `n_estimators` via grid or randomised search.

Pipeline Pseudocode

A high-level view of the pipeline logic (no confidential code or data).

```
INPUT: train.csv, test.csv    # each row is a record pair: text1, text2, label
                                # label = 1 if same person, else 0
                                epsilon                # privacy budget (smaller = more privacy)

# 1. Turn text into numbers
features = TF-IDF(text1, text2)          # weight informative words
features = TruncatedSVD(features, 100)   # shrink to 100 numbers per field

# 2. Model A: Raw baseline (no privacy)
model_raw = XGBoost().train(features_train, labels_train)
evaluate(model_raw, test)

# 3. Model B: LDP (privacy by adding noise)
flip_prob = 1 / (1 + e^(epsilon / 2))    # smaller epsilon -> more noise
noisy = features + random_noise(flip_prob)
model_ldp = XGBoost().train(noisy_train, labels_train)
evaluate(model_ldp, test)

# 4. Model C: Bloom filter (privacy by hashing)
bits = encode each record into a Bloom filter
bits = add_privacy_noise(bits)            # flip some bits
for each pair of records in the same block:
    feature = Dice_similarity(bits_i, bits_j)
    label   = (same person?)
```

```
# handle imbalance: keep all matches, keep 3x as many non-matches
balanced = downsample_non_matches(ratio = 3 : 1)
split balanced -> 70% train / 30% test
model_bf = XGBoost().train(bf_train, bf_labels_train)
evaluate(model_bf, test)
```

```
# 5. Compare all three on Accuracy / Precision / Recall / F1
```

Tools & Skills

Python, XGBoost, scikit-learn (TF-IDF, Truncated SVD), Bloom filter encoding, Local Differential Privacy, React (dashboard integration), Agile / GitHub / Kanban.

Confidentiality

Source code, algorithms, and the sponsor datasets remain confidential to the university research project. This document describes the methodology and presents non-confidential results and synthetic-data demonstrations only.